Version 2.5 Update

Flux-Preserving Adaptive Finite State Projection
for Multiscale Stochastic Reaction Networks

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April 13, 2026

Dear Editor,

After submission of the revised manuscript, we identified two issues in the submitted files and corrected them in Version 2.5. These corrections do not change the implementation, the numerical experiments, or the conclusions of the paper. They are intended only to ensure that the manuscript and response letter accurately reflect the mathematical argument and the benchmark parameters used in the submitted experiments.

Version 2.5 Corrections

Correction 1: Adaptive time-step proposition (Section 3).

In the submitted version, Proposition 3.5 used language that was stronger than what was established by the proof. In particular, the proof supports an upper bound on the leading-order *total variation change* of the compressed distribution over one step, rather than a stronger interpretation in terms of exact probability redistribution.

We therefore revised the proposition and proof so that they now state and prove the bound

$$\frac{1}{2}\,\|\widehat{\boldsymbol{p}}(t_n + \Delta t_n) - \widetilde{\boldsymbol{p}}_n\|_1 \leq \Phi_{\text{total}}(J_n, t_n)\,\Delta t_n + O(\Delta t_n^2).$$

where $\widehat{\boldsymbol{p}}$ denotes the compressed dynamics on the active set. The proof now proceeds by bounding

$$\|\widetilde{\mathbf{A}}_{JJ}\widetilde{\boldsymbol{p}}_n\|_1 \leq 2\,\Phi_{\text{total}}(J_n, t_n),$$

which is the rigorous statement needed for the argument. Accordingly, the step rule

$$\Delta t_n = \varepsilon_{\Delta t}/\Phi_{\text{total}}(J_n, t_n)$$

is now described as a temporal-resolution criterion for the compressed dynamics, which is what the proof justifies. We also updated the corresponding discussion in the response letter to match this corrected statement.

Correction 2: Bottleneck parameter consistency (Section 5).

In the bottleneck benchmark, the submitted manuscript contained an inconsistent textual value for the downstream rate constant. The experiment uses

$$k_2 = 0.1\,\mathrm{s}^{-1},$$

and Version 2.5 corrects the corresponding text accordingly. This change is purely textual and does not affect the experiment, numerical results, or conclusions.

Summary.

Version 2.5 therefore makes two targeted corrections:

1. a proof-level correction to Proposition 3.5 and the matching explanation in the response letter, and

2. a parameter-consistency correction in the bottleneck benchmark discussion.

No code, numerical result, or scientific conclusion has changed.